

Lecture 12a

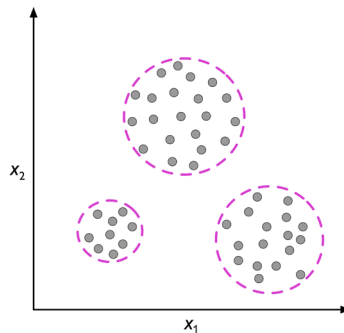
Unsupervised Learning

Unsupervised Learning

No labels!

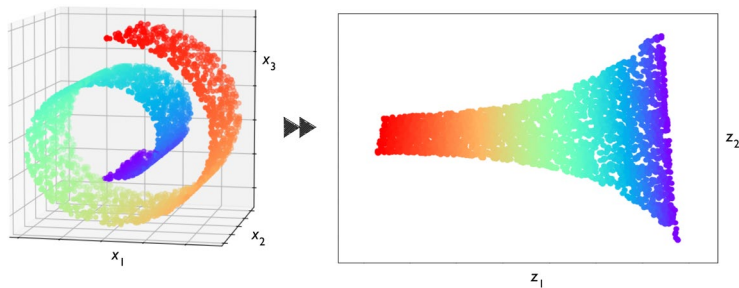
1. Clustering

K-means

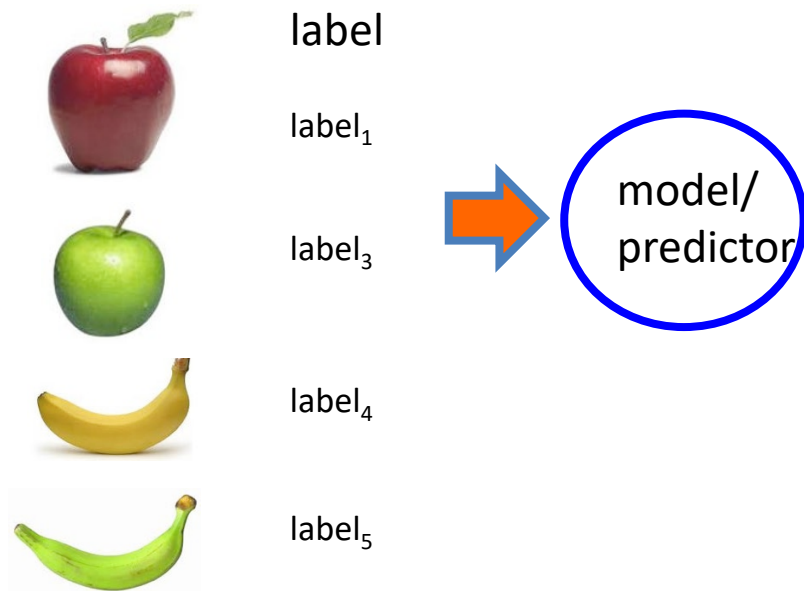


2. Dimensionality reduction

PCA

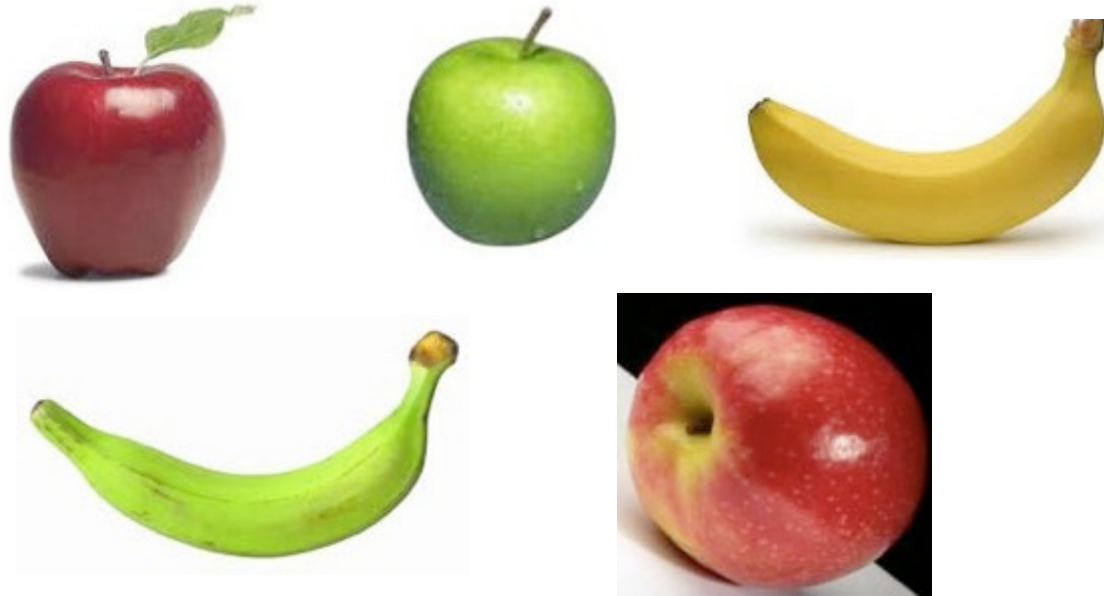


Supervised learning



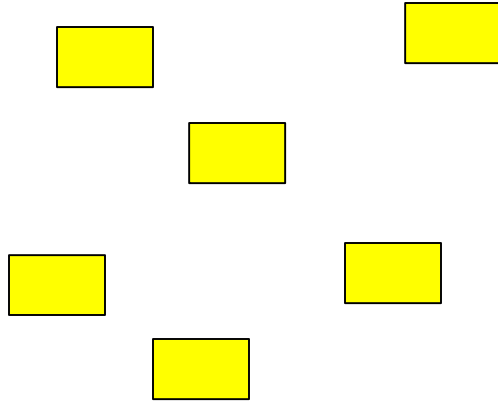
Supervised learning: given labeled examples

Unsupervised learning



Unsupervised learning: given data, i.e. examples, but no labels

Unsupervised learning



Given some example without labels, do something!

Unsupervised learning applications

learn clusters/groups without any label

customer segmentation (i.e. grouping)

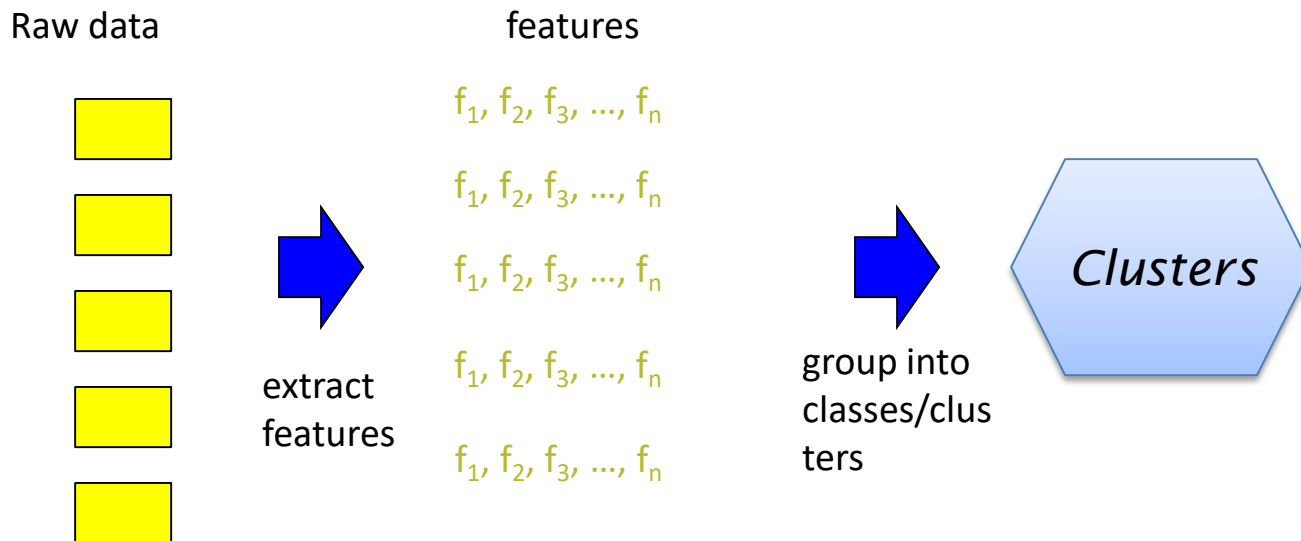
image compression

bioinformatics: learn motifs

find important features

...

Unsupervised learning: clustering



No "supervision", we're only given data and want to find natural groupings

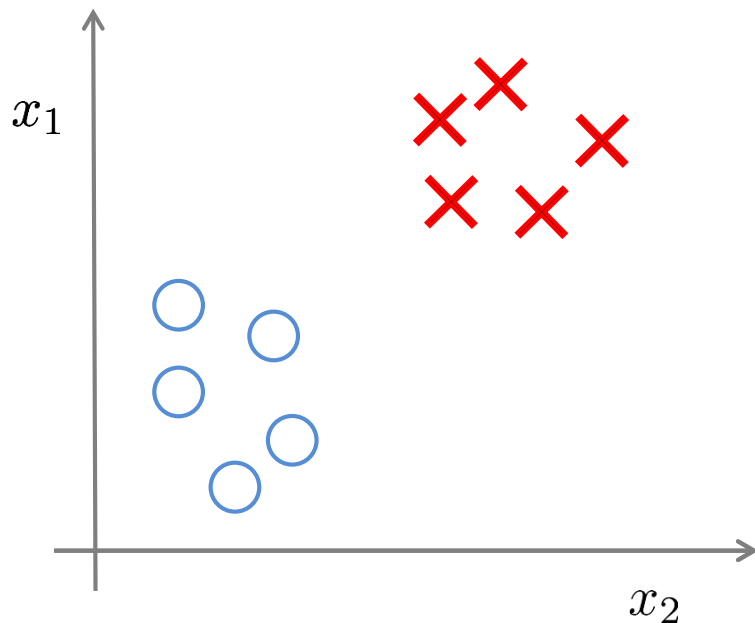
Unsupervised learning: modeling

Most frequently, when people think of unsupervised learning they think clustering

Another category: learning probabilities/parameters for models without supervision

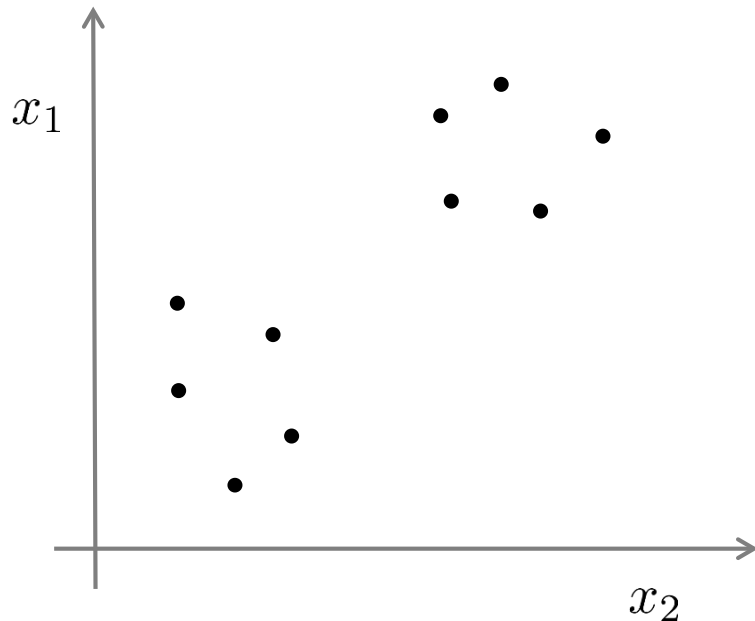
- Learn a translation dictionary
- Learn a grammar for a language
- Learn the social graph

Supervised learning



Training set: $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), (x^{(3)}, y^{(3)}), \dots, (x^{(m)}, y^{(m)})\}$

Unsupervised learning



Training set: $\{x^{(1)}, x^{(2)}, x^{(3)}, \dots, x^{(m)}\}$

Clustering

What is Clustering?

- Cluster: a collection of data objects
 - Similar to one another within the same cluster
 - Dissimilar to the objects in other clusters
- Cluster analysis
 - Grouping a set of data objects into clusters
- Clustering is **unsupervised classification**:
 - no predefined classes
- Typical applications
 - As a **stand-alone tool** to get insight into data distribution
 - As a **preprocessing step** for other algorithms

Examples of Clustering Applications

- Marketing: Help marketers discover distinct groups in their customer bases, and then use this knowledge to develop targeted marketing programs
- Land use: Identification of areas of similar land use in an earth observation database
- Insurance: Identifying groups of motor insurance policy holders with a high average claim cost
- Urban planning: Identifying groups of houses according to their house type, value, and geographical location
- Seismology: Observed earth quake epicenters should be clustered along continent faults

What Is a Good Clustering?

- A good clustering method will produce clusters with
 - High intra-class similarity
 - Low inter-class similarity
- Precise definition of clustering quality is difficult
 - Application-dependent
 - Ultimately subjective

Data Requirements for Clustering

- Scalability
- Ability to deal with different types of attributes
- Discovery of clusters with arbitrary shape
- Minimal domain knowledge required to determine input parameters
- Ability to deal with noise and outliers
- Insensitivity to order of input records
- Robustness wrt high dimensionality
- Incorporation of user-specified constraints
- Interpretability and usability

Major Clustering Approaches

- Partitioning: Construct various partitions and then evaluate them by some criterion
- Hierarchical: Create a hierarchical decomposition of the set of objects using some criterion
- Model-based: Hypothesize a model for each cluster and find best fit of models to data
- Density-based: Guided by connectivity and density functions

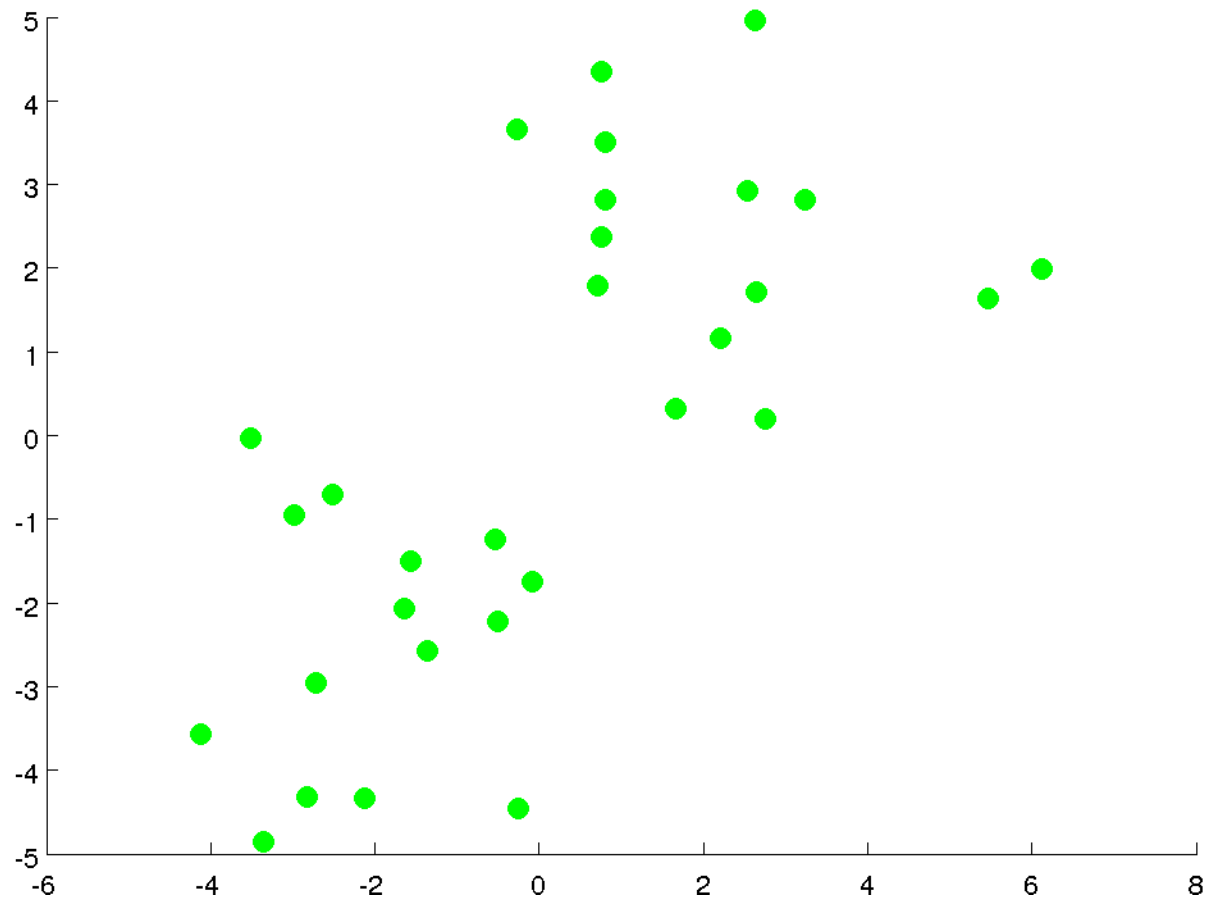
Partitioning Algorithms

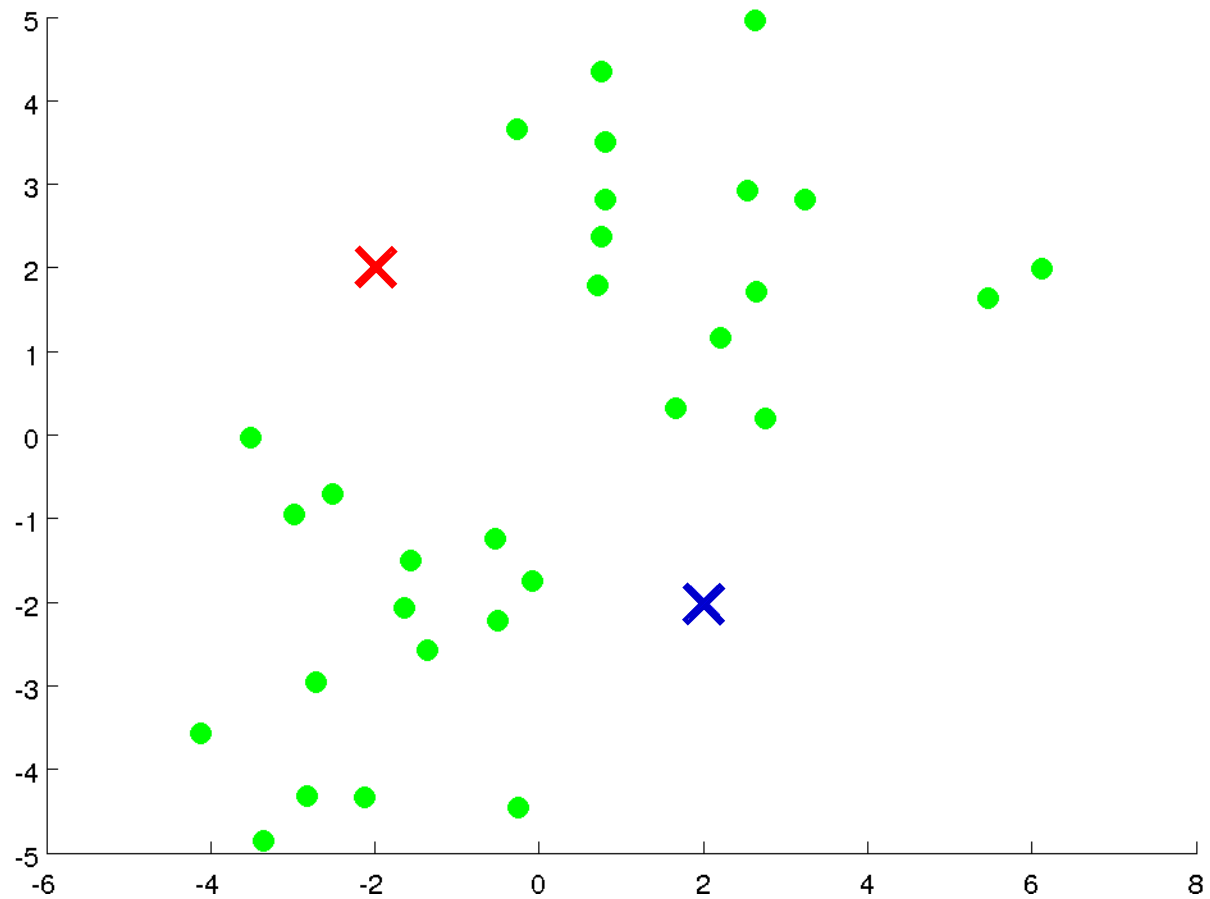
- Partitioning method: Construct a partition of a database D of n objects into a set of k clusters
- Given a k , find a partition of k clusters that optimizes the chosen partitioning criterion
 - Global optimal: exhaustively enumerate all partitions
 - Heuristic methods: *k-means* and *k-medoids* algorithms
 - k-means (MacQueen, 1967): Each cluster is represented by the center of the cluster
 - k-medoids or PAM (Partition around medoids) (Kaufman & Rousseeuw, 1987): Each cluster is represented by one of the objects in the cluster

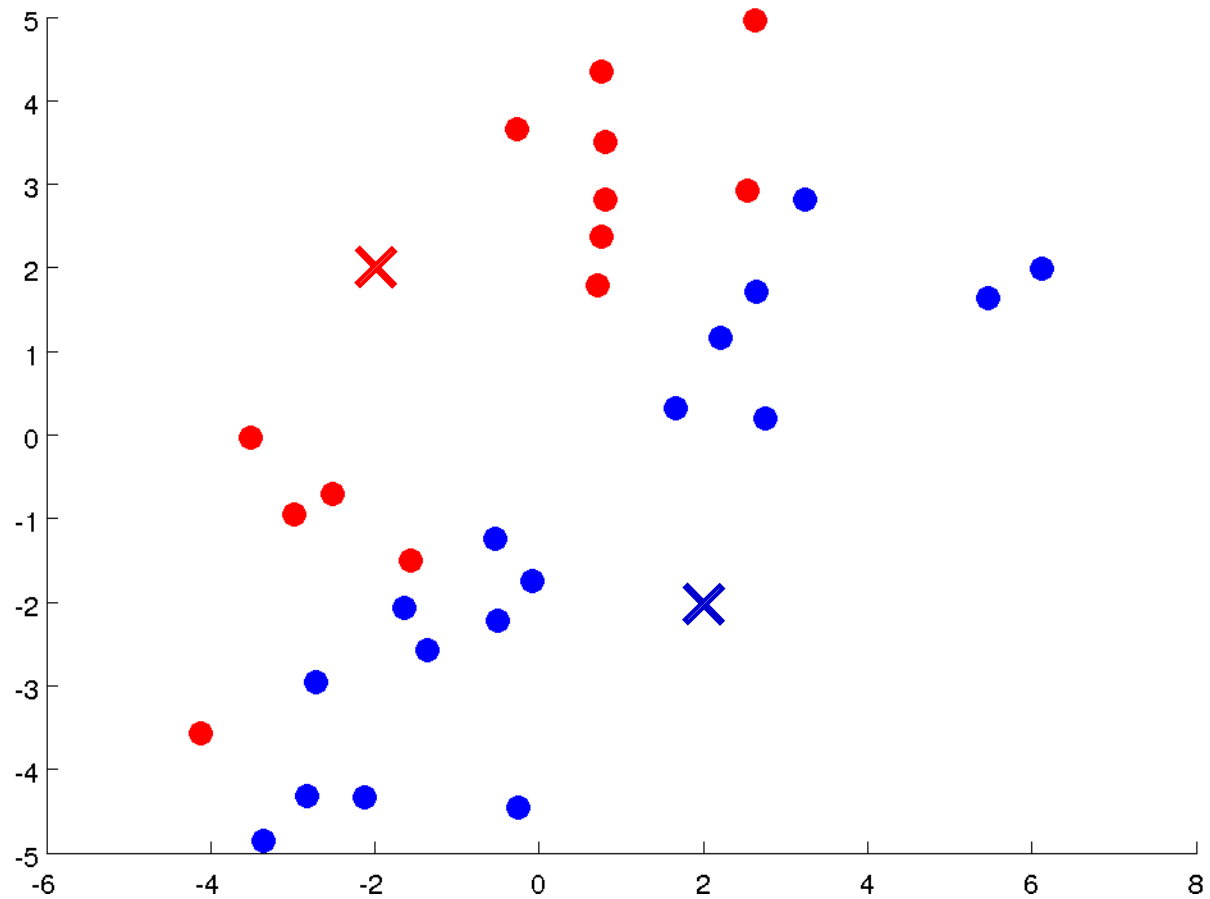
K-Means Clustering

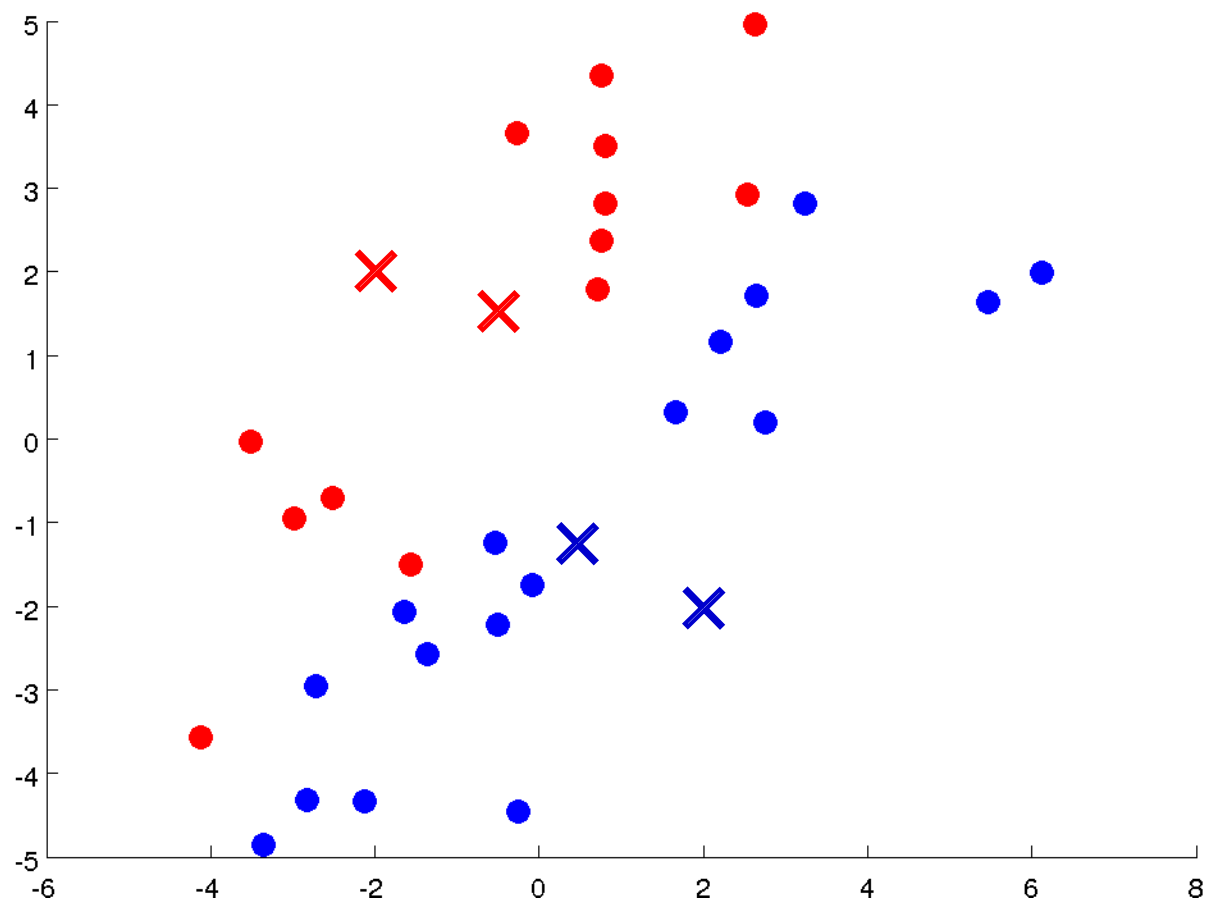
- Given k , the *k-means* algorithm consists of four steps:
 - Select initial centroids at random.
 - Assign each object to the cluster with the nearest centroid.
 - Compute each centroid as the mean of the objects assigned to it.
 - Repeat previous 2 steps until no change.

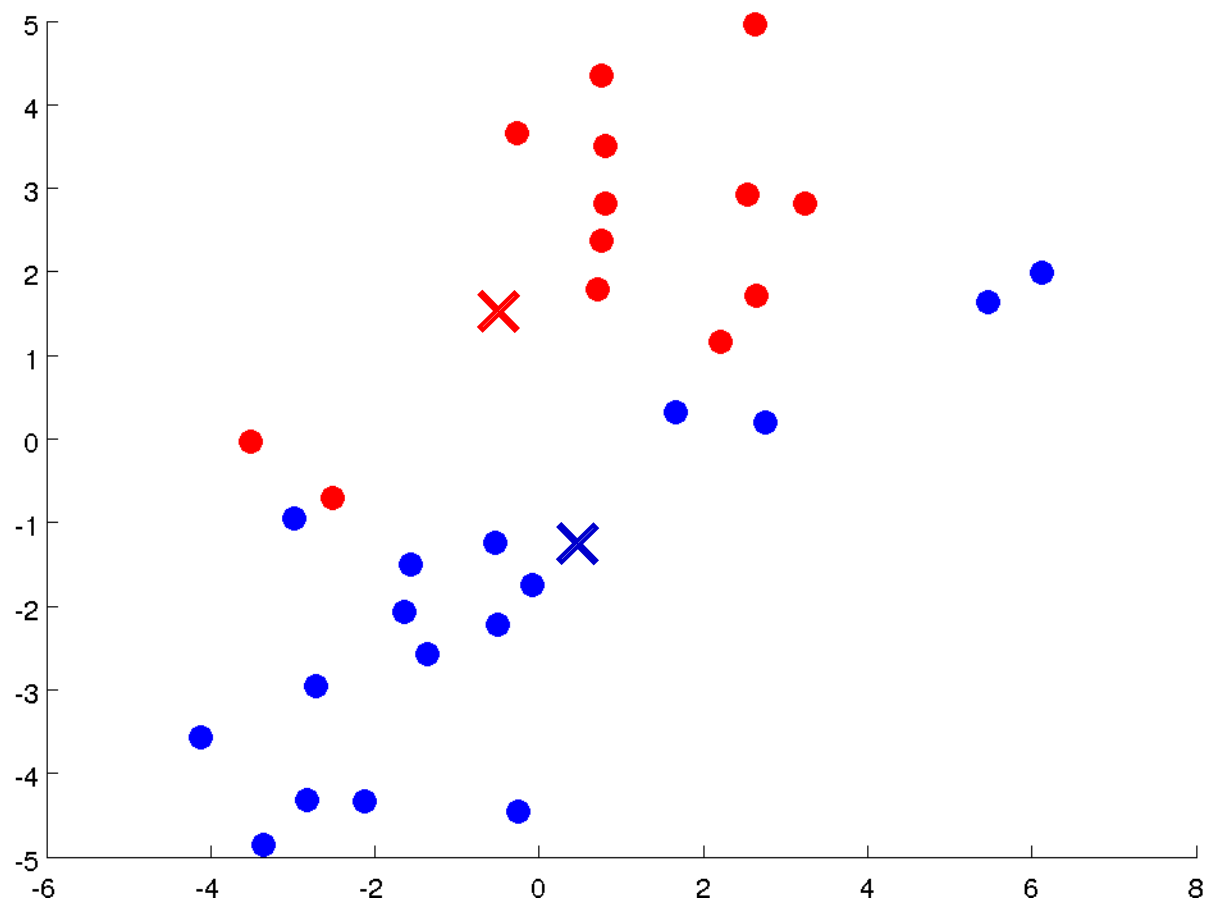
K-means clustering

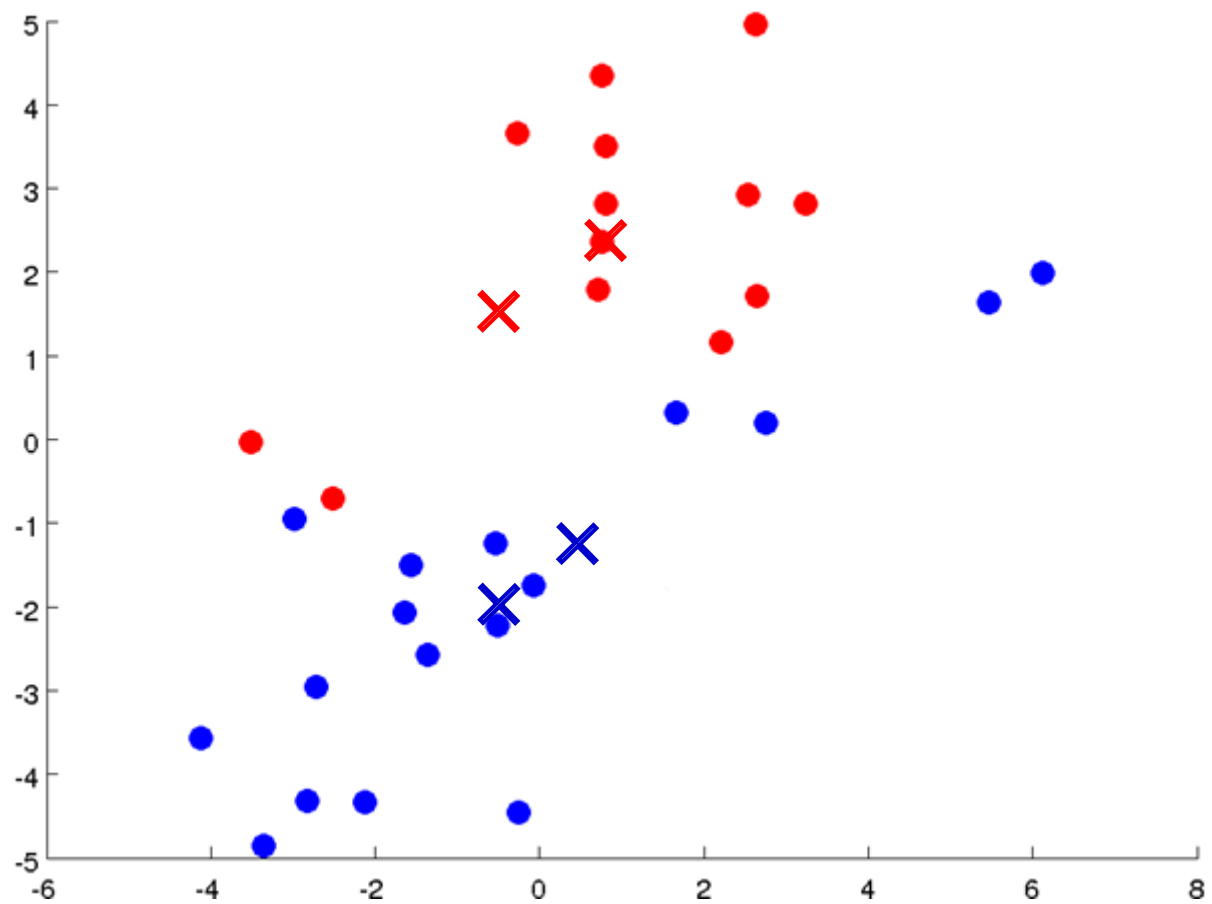


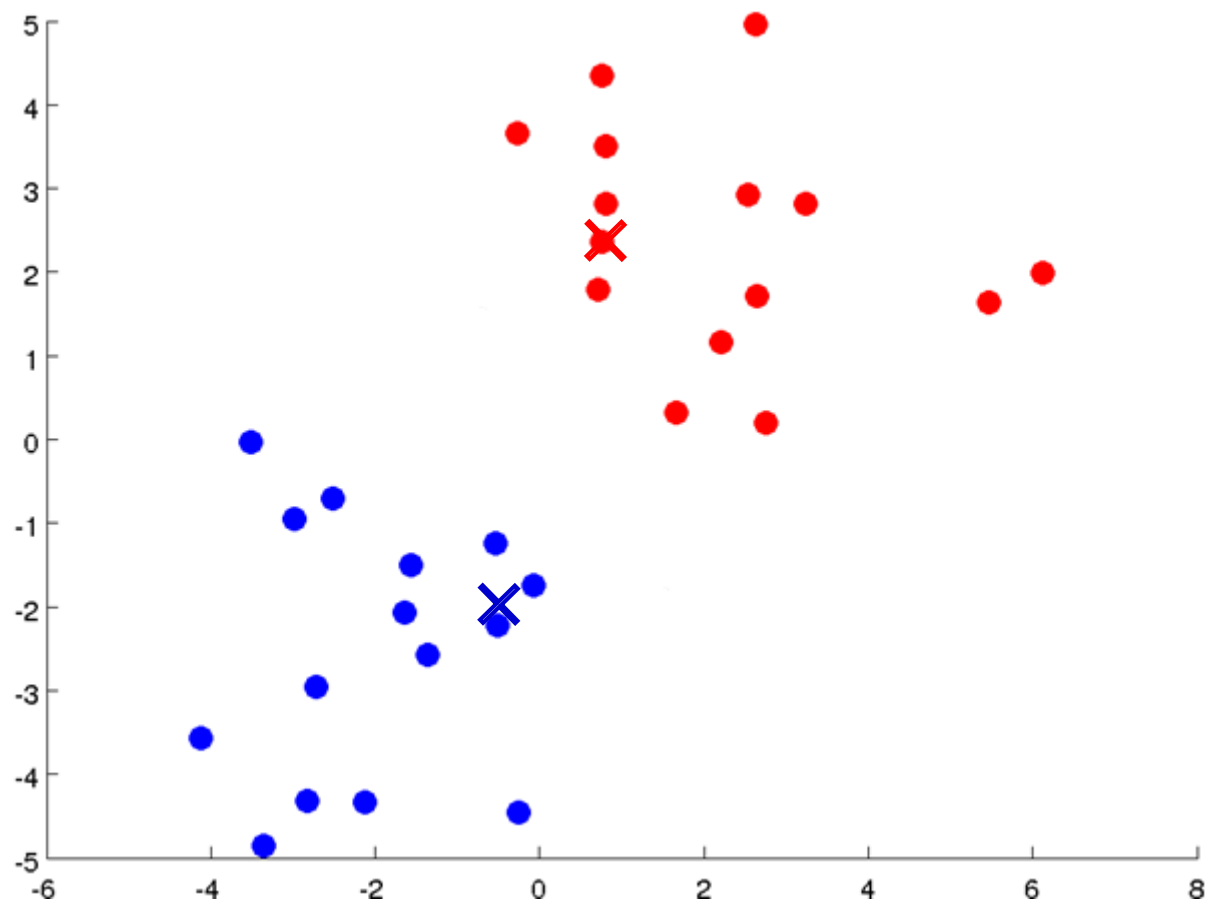


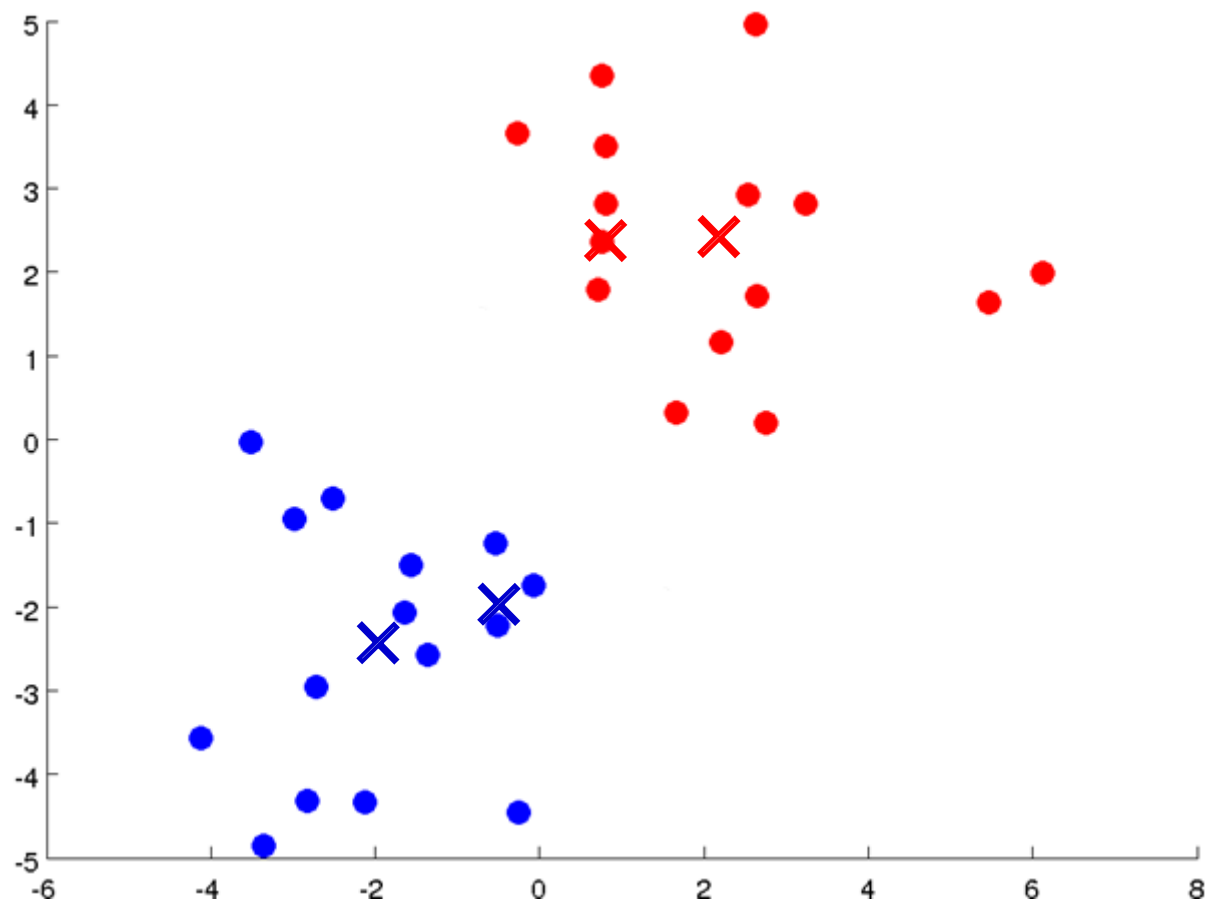


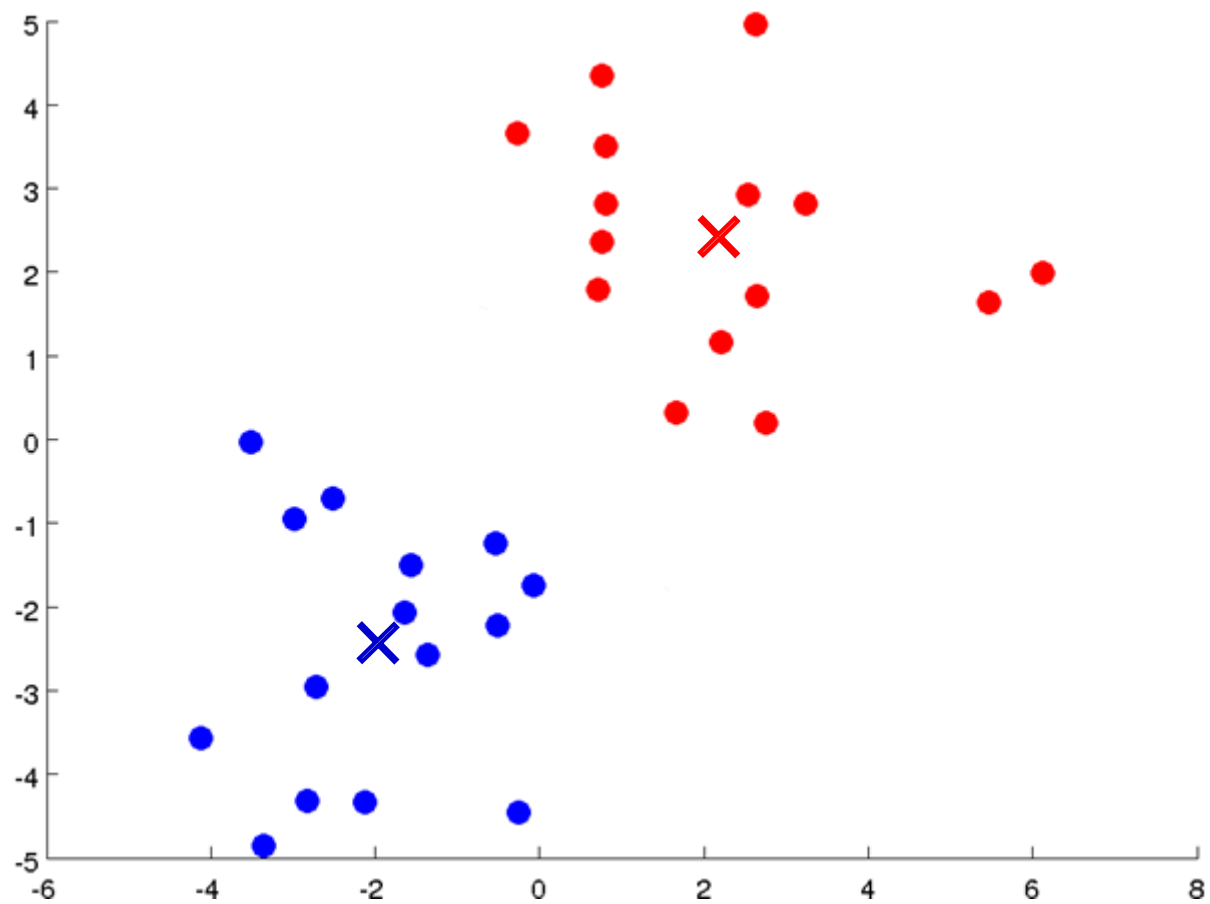












K-means algorithm

Input:

- K (number of clusters)
- Training set $\{x^{(1)}, x^{(2)}, \dots, x^{(m)}\}$

$x^{(i)} \in \mathbb{R}^n$ (drop $x_0 = 1$ convention)

K-means algorithm

Randomly initialize K cluster centroids $\mu_1, \mu_2, \dots, \mu_K \in \mathbb{R}^n$

Repeat {

 for $i = 1$ to m

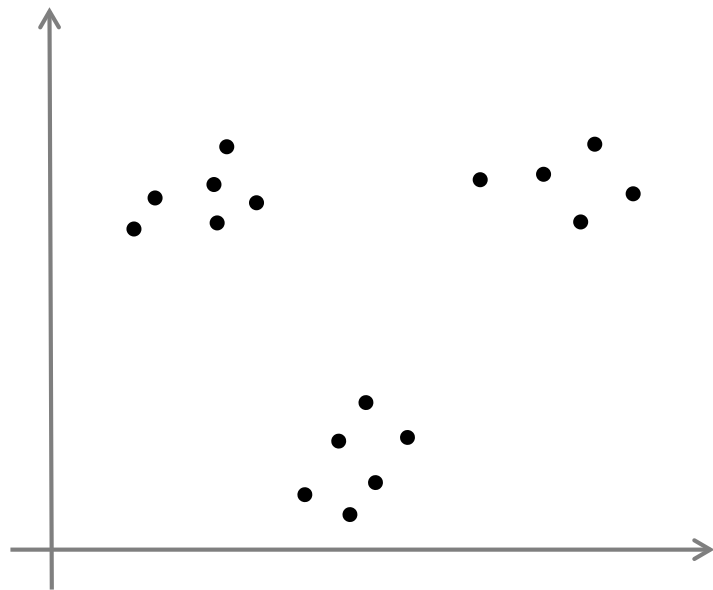
$c^{(i)} :=$ index (from 1 to K) of cluster centroid
 closest to $x^{(i)}$

 for $k = 1$ to K

$\mu_k :=$ average (mean) of points assigned to cluster k

}

K-means for non-separated clusters



K-means optimization objective

$c^{(i)}$ = index of cluster $(1, 2, \dots, K)$ to which example $x^{(i)}$ is currently assigned

μ_k = cluster centroid k ($\mu_k \in \mathbb{R}^n$)

$\mu_{c^{(i)}}$ = cluster centroid of cluster to which example $x^{(i)}$ has been assigned

Optimization objective:

$$J(c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_K) = \frac{1}{m} \sum_{i=1}^m \|x^{(i)} - \mu_{c^{(i)}}\|^2$$

$$\min_{\substack{c^{(1)}, \dots, c^{(m)}, \\ \mu_1, \dots, \mu_K}} J(c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_K)$$

K-means algorithm

Randomly initialize K cluster centroids $\mu_1, \mu_2, \dots, \mu_K \in \mathbb{R}^n$

Repeat {

 for $i = 1$ to m

$c^{(i)} :=$ index (from 1 to K) of cluster centroid
 closest to $x^{(i)}$

 for $k = 1$ to K

$\mu_k :=$ average (mean) of points assigned to cluster k

}

K-means Initialization

K-means algorithm

Randomly initialize K cluster centroids $\mu_1, \mu_2, \dots, \mu_K \in \mathbb{R}^n$

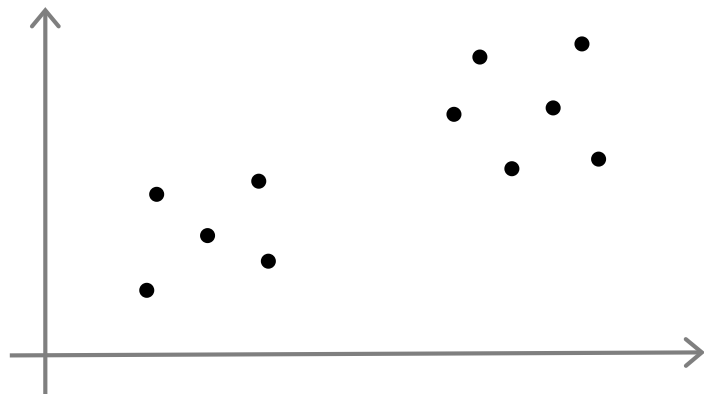
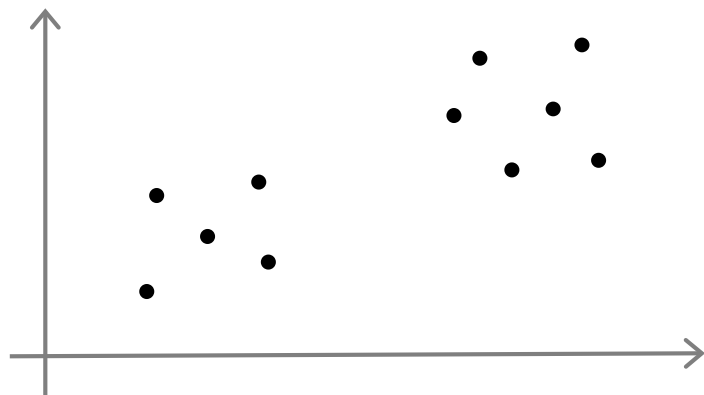
Repeat {
 for $i = 1$ to m
 $c^{(i)} :=$ index (from 1 to K) of cluster centroid
 closest to $x^{(i)}$
 for $k = 1$ to K
 $\mu_k :=$ average (mean) of points assigned to cluster k
}

Random initialization

Should have $K < m$

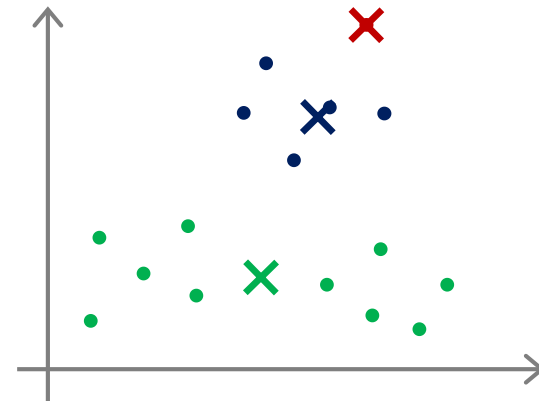
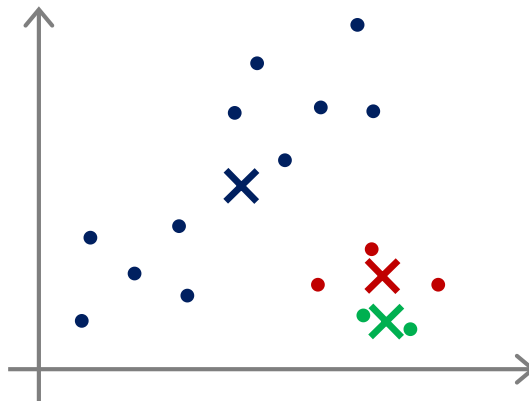
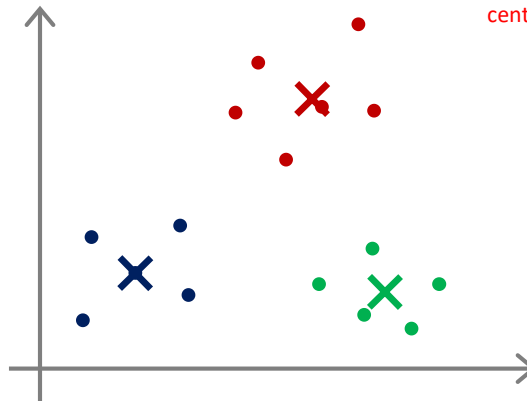
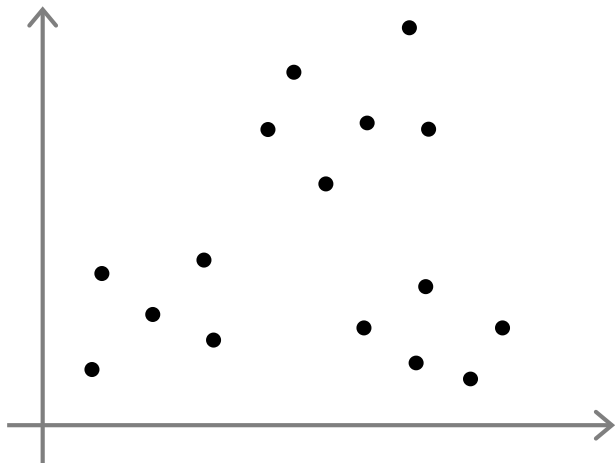
Randomly pick K training examples.

Set $\mu_1, \dots, \mu_K$ equal to these K examples.



Local optima

Depending on the initialization of cluster centroids K-means can produce different results



Random initialization

For $i = 1$ to 100 {

 Randomly initialize K-means.

 Run K-means. Get $c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_K$.

 Compute cost function (distortion)

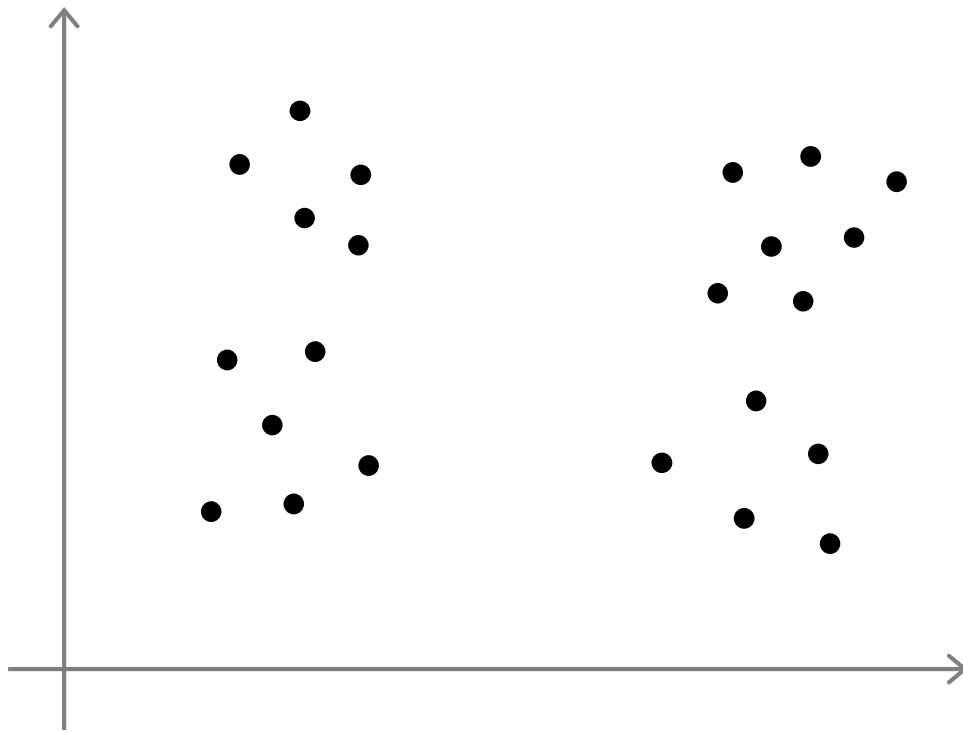
$$J(c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_K)$$

}

Pick clustering that gave lowest cost $J(c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_K)$

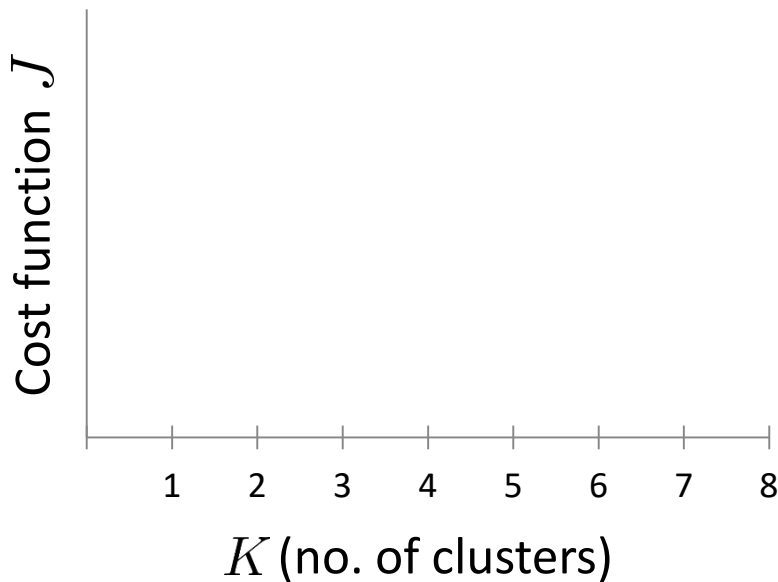
K-means Number of Clusters

What is the right value of K?

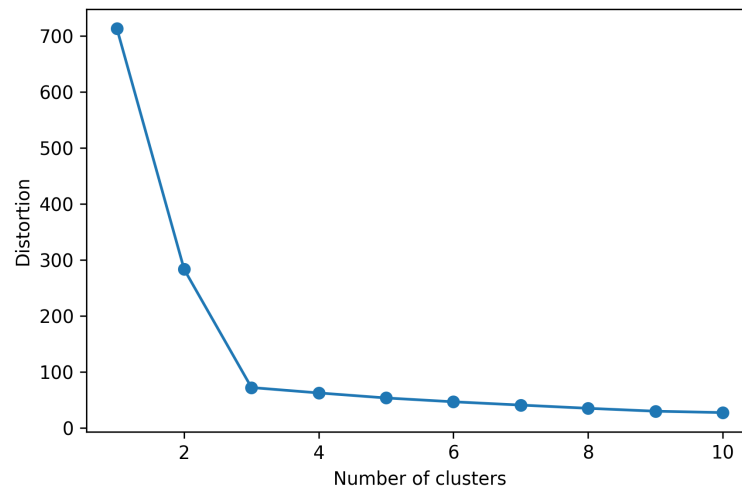


Choosing the value of K empirically

Elbow method:



We use within-cluster **SSE** to measure the model distortion. Intuitively, the larger K we choose, the smaller SSE the model should have.



Draw the distortion for different K , we can pick $K=3$ empirically.

K-means ++

K-means

Basically, we have four steps for k-means:

Step 1: Randomly choose k centroids as initial center for each class.

Step 2: Pick one sample from all unclassified data points, and assign it into the class with the nearest center.

Step 3: Re-calculate the center of the class with the new sample.

Step 4: Stop based on convergence criteria

- No change in clusters
- Max iterations

K-means

What is nearest? How to define distance?

Like KNN, we still use **squared Euclidean distance**:

$$d(\mathbf{x}, \mathbf{y})^2 = \sum_{j=1}^m (x_j - y_j)^2 = \|\mathbf{x} - \mathbf{y}\|_2^2$$

Based on this Euclidean distance metric, we can describe the k-means algorithm as a simple optimization problem, an iterative approach for minimizing the within-cluster **Sum of Squared Errors (SSE)**, which is sometimes also called **cluster inertia**:

$$SSE = \sum_{i=1}^n \sum_{j=1}^k w^{(i,j)} \|\mathbf{x}^{(i)} - \boldsymbol{\mu}^{(j)}\|_2^2$$

Here $\boldsymbol{\mu}^{(j)}$ is the representative point (centroid) for cluster j , and $w^{(i,j)} = 1$ if the sample $\mathbf{x}^{(i)}$ is in cluster j ; $w^{(i,j)} = 0$ otherwise.

K-means

- Strengths
 - Simple iterative method
 - User provides “K”
- Weaknesses
 - Very sensitive to initial centroids
 - Difficult to guess the correct “K”

So, we need a better centroids initialization method.

K-means++

Centroids finding:

1. Initialize an empty set $\mathbf{M}$ to store the k centroids being selected.
2. Randomly choose the first centroid $\mu^{(j)}$ from the input samples and assign it to $\mathbf{M}$.
3. For each sample $\mathbf{x}^{(i)}$ that is not in $\mathbf{M}$, find the minimum squared distance $d(\mathbf{x}^{(i)}, \mathbf{M})^2$ to any of the centroids in $\mathbf{M}$.
4. To randomly select the next centroid $\mu^{(p)}$, use a weighted probability distribution equal to
$$\frac{d(\mu^{(p)}, \mathbf{M})^2}{\sum_i d(\mathbf{x}^{(i)}, \mathbf{M})^2}.$$
5. Repeat steps 2 and 3 until k centroids are chosen.
6. Proceed with the classic k-means algorithm.

K-means ++

1. Choose one center uniformly at random among the data points.
2. For each data point x not chosen yet, compute $D(x)$, the distance between x and the nearest center that has already been chosen.
3. Choose one new data point at random as a new center, using a weighted probability distribution where a point x is chosen with probability proportional to $D(x)^2$.
4. Repeat Steps 2 and 3 until k centers have been chosen.
5. Now that the initial centers have been chosen, proceed using standard k -means clustering.

D. Arthur and S. Vassilvitskii, "k-means++: The Advantages of Careful Seeding" (2007)

<http://ilpubs.stanford.edu:8090/778/1/2006-13.pdf>